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```
import pandas as pd
```

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

```
data = {  
    "name": ["Sally", "Mary", "John"],  
    "age": [50, 40, 30],  
    "qualified": [True, False, False]  
}  
df = pd.DataFrame(data)  
print(df)  
print("Priyanshi Gupta")
```

	name	age	qualified
0	Sally	50	True
1	Mary	40	False
2	John	30	False

Priyanshi Gupta

5b. Write a python code to show statistical information on given data set.

```
import statistics
```

Mean

```
# Calculate average values  
print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))  
print(statistics.mean([1, 3, 5, 7, 9, 11]))
```

```
print(statistics.mean([-11, 5.5, -3.4, 7.1, -9, 22]))
print("Priyanshi Gupta")
```

```
7
6
1.8666666666666667
Priyanshi Gupta
```

Median

```
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median([1, 3, 5, 7, 9, 11]))
print(statistics.median([-11, 5.5, -3.4, 7.1, -9, 22]))
print("Priyanshi Gupta")
```

```
7
6.0
1.05
Priyanshi Gupta
```

Mode

```
import statistics

# Calculate the mode
print(statistics.mode([1, 3, 3, 3, 5, 7,7,9, 11]))
print(statistics.mode([1, 1, 3, -5, 7, -9, 11]))
print(statistics.mode(['red', 'green', 'blue', 'red']))
print("Priyanshi Gupta")
```

```
3
1
red
Priyanshi Gupta
```

Standard deviation

```
print(statistics.stdev([1, 3, 5, 7, 9, 11]))
print(statistics.stdev([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.stdev([-11, 5.5, -3.4, 7.1]))
print(statistics.stdev([1, 30, 50, 100]))
print("Priyanshi Gupta")
```

```
3.7416573867739413
0.6925797186365383
8.414471660973927
41.67633221226007
Priyanshi Gupta
```

Variance

```
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print(statistics.variance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.variance([-11, 5.5, -3.4, 7.1]))
print(statistics.variance([1, 30, 50, 100]))
print("Priyanshi Gupta")
```

```
14
0.47966666666666663
70.80333333333333
1736.9166666666667
Priyanshi Gupta
```

5c. Write a python code to create pandas series from dictionaries.

```
# create a dictionary
dictionary = {'A': 10, 'B': 20, 'C': 30}

# create a series
```

```
series = pd.Series(dictionary)
```

```
print(series)  
print("Priyanshi Gupta")
```

```
A    10  
B    20  
C    30  
dtype: int64  
Priyanshi Gupta
```

```
# create a dictionary  
dictionary = {'A': 50, 'B': 10, 'C': 80}  
  
# create a series  
series = pd.Series(dictionary, index=['B', 'C', 'A'])  
  
print(series)  
print("Priyanshi Gupta")
```

```
B    10  
C    80  
A    50  
dtype: int64  
Priyanshi Gupta
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

```
# dictionary of lists  
dict = {'name': ["aparna", "pankaj", "sudhir", "Geeku"],  
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],  
        'score': [90, 40, 80, 98]}  
  
df = pd.DataFrame(dict, index = [True, False, True, False])
```

```
print(df)
print("Priyanshi Gupta")
```

	name	degree	score
True	aparna	MBA	90
False	pankaj	BCA	40
True	sudhir	M.Tech	80
False	Geeku	MBA	98
Priyanshi Gupta			